Variant evaluation table: Bike Sharing linear regression

Rules

- Change ONE thing per row. Judge every row on the validation slice, never on test.
- For log-target rows, convert predictions back to rentals first. Compare RMSE and MAE in rentals, not R^2.
- Write your bet before you run the variant. If you did not, write "no bet". Do not fill it in afterwards.
- Decide your keep rule before you look (for example: keep a group only if validation RMSE improves by more than 1%).
- Optional: note the condition number of X^TX in the Note column.

ID	What changed (one thing only)	Features and count	Target	Train R^2	Val R^2	Val RMSE (rentals)	Val MAE (rentals)	Δ RMSE vs previous	Key weight (SE, t)		Hit / Miss	Note: what surprised me
V0	Starting with 3 continuous features only.	feats = ['atemp', 'hum', 'windspeed']	‘cnt’	0.277								
V1	Added “hr” feature	features = ['hr', 'atemp', 'hum', 'windspeed']	‘cnt’	0.321	0.203	193.08	138.97	0	Atemp:[6.51, 44.98] Hr: [0.168, 36.45] Hum: [5.97, -25.41]			Humidities negative t-stats, it happened as weight to hum is negative.
V2	Cyclic sin and cos encoded “hr” feature	features = ['hr', 'atemp', 'hum', 'windspeed']	‘cnt’	0.429	0.279	184.22	127.36	+11.61 difference	Atemp;[6.10, 39.05], Hr_sin:[1.5, -43.29] Hr_cos:[1.52, -47.80]			Cyclic encoding gave negative weight t hr_sin, hr_cos.
V3	“OneHotEncoded” “hr” feature	features = ['hr', 'atemp', 'hum', 'windspeed']	‘cnt’	0.616	0.485	155.73	110.03	+28.49 difference				One hot encoded “hr” was better for our model than cyclic encoded.
V4	Added “season”, dropped “hr”, encoded “season”	features = ["season", "atemp", "hum", "windspeed"]	‘cnt’	0.259	0.150	200.03	149.14	-44.3 difference				Adding season dropped our models was less affected by seasons as per t-stats.
V5	Added “hr” along with “season”, both un-encoded	features = ["season", "hr" , "atemp", "hum", "windspeed"]	‘cnt’	0.323	0.203	193.62	139.14	+6.41				Not encoding our features drops model’s t recognition power
V6	Now encoding hr and keeping season same	features = ["season", "hr" , "atemp", "hum", "windspeed"]	‘cnt’	0.617	0.480	156.41	110.46	+37.21				Encoding hr is powerful and helps our model to understand more from encoded hr feature

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V7	Encoding “season” and keeping “hr” same	features = ["season", "hr" , "atemp", "hum", "windspeed"]	‘cnt’	0.333	0.218	191.81	137.36	-35.4				Hr is still a great feature for our model
V8	Encoded both “hr” and “season” now	features = ["season", "hr" , "atemp", "hum", "windspeed"]	‘cnt’	0.623	0.498	153.92	108.57	37.89				This was the lowest rmse and highest val r2 yet.
V9	Dropped season and hr, added raw weathersit,	features = ["weathersit", "atemp", "hum", "windspeed"]	‘cnt’	0.240	0.136	201.69	151.51	-47.77				weather situation feature weakened our model, as we can see from t-stats
V10	Added hr and season back, along with weathersit, raw nothing encoded	features = ["season", "hr" ,"weathersit", "atemp", "hum", "windspeed"]	‘cnt’	0.322	0.203	193.65	139.67	8.04				Now t-stats for weathersir has increase I think due to hr and season, mainly season
V11	Encoding hr, season and weathersit	features = ["season", "hr" , "weathersit" , "atemp", "hum", "windspeed"]	‘cnt’	0.632	0.499	153.58	108.05	40.07				Hr,season and weather are helping our model to get more better, highest val r2 and train r2
V12	Including every feature, except temp, raw and unencoded	features = ["holiday", "yr", "workingday", "season","weathersit", "hr", "atemp", "hum", "windspeed"]	‘cnt’	0.363	0.286	183.24	146.77	-29.66				Added holiday, year and workingday for day wise features
V13	Encoding “hr”, “season” and “weathersit”	features = ["holiday", "yr", "workingday", "season","weathersit", "hr", "atemp", "hum", "windspeed"]	‘cnt’	0.675	0.605	136.33	107.84	46.91				Encoding has lead us straight to the highest r2 and lowest RMSE ever, with these many features
V14	Now logging ‘cnt’ and keeping everything same	features = ["holiday", "yr", "workingday", "season","weathersit", "hr", "atemp", "hum", "windspeed"]	‘cnt’ Logged	0.814	0.806	0.60 logged, 146.66	0.46 logged, 99.90	-10.33				Logging our target variable recused it’s skewness, BEST EVER Val and Train r2. Near perfect midnight and early morning predictiosn

ID	What changed (one thing only)	Features and count	Target	Train R^2	Val R^2	Val RMSE (rentals)	Val MAE (rentals)	Δ RMSE vs previous	Key weight (SE, t)		Hit / Miss	Note: what surprised me
V15	Including temp and dropping “atemp”, everything same and encoded. Y logged	features = ["holiday", "yr", "season", "weathersit", "hr", "temp", "hum", "windspeed"]	‘cnt’ logged	0.814	0.803	0.606 logged, 153.18	0.466 logged, 103.57	-6.67				As we know frm EDA, temp and atemp are correlated with 0.998 value, so they lie in same vector space, rmse increased but r2 is same.
V16	Including both temp and atemp, let us see how 0.998 correlation affects our model	features = ["holiday", "yr", "season", "weathersit", "hr", "temp","atemp", "hum", "windspeed"]	‘cnt’ logged	0.814	0.804	149.88	101.71	3.33				Shocked to see that multicollinearity has no effect on metrics, it has effect on weight allocation and feature importance, seen via beta_hat and t-stats also standard error, all of them got split as our mode don’t know whom to give credit for relationship.

My keep rule (decided before looking): _____

Model I would ship, and why: _____

Compression line ("X is just Y with Z changed"): _____